PoseSplit: A Synthetic-to-Real Audio-Plus-Pose Pilot for Camera Reframing

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Camera reframing and microphone reorientation change different relationships between a scene and its sensors. We report a frozen CPU pilot that separates these operations in a small synthetic-to-real spatial task. Five seeds produce 1,600 four-channel waveforms: 1,280 for training and 320 for validation. Fifteen small predictors use acoustic intensity, coherence and known relative yaw to classify geometric field-of-view membership. Evaluation uses 168 half-second windows from 14 real STARSS23 recordings in seven held-out rooms, with four fixed headings. Under camera-only reframing, mean balanced accuracy is .829 for pose-aware training, .480 for canonical-orientation training, .408 for intentionally incorrect pose labels and .782 for analytic geometry. Under virtual co-rig reorientation, canonical training instead reaches .841 versus .832 for pose-aware training. Thus pose-aware training does not dominate both regimes. The analytic direction estimate has median error 23.48 degrees, compared with 50.14 for a constant-front reference; this is a separate measurement from the learned binary task. Independent raw-data replay verifies the reported observations. The results support a bounded audio-plus-known-pose data interface and real-room feasibility. They do not demonstrate visual grounding, realistic neural audio generation, unseen-angle generalization, or the proposed development-driven coverage sampler.

Additional Key Words and Phrases: synthetic data, spatial audio, sensor pose, geometric field of view, synthetic-to-real evaluation

1 Introduction and Background

A camera may turn toward a doorway while its microphone remains fixed. The recorded sound then stays unchanged, although the source's position relative to the image changes. Rotating the complete rig is different: the camera view and microphone-relative sound direction both change. A dataset constructed only for one relationship can leave a predictor without support for the other. Before training a large generator, a small experiment can establish whether the signal, coordinate conventions and real-data evaluation are usable.

Spatial synthesis is established prior work. SELDVisualSynth creates moving visual objects aligned with synthetic spatial sound [7]. The official DCASE generator and SAVGBench rotate spatial audio and extract perspective video using a common viewing direction [9, 10]. ViSAGe already conditions spatial audio on camera direction; its later extension studies scene-aware generation and efficiency [3, 4]. PhysWave uses trajectory conditioning and acoustic priors [11], StereoFoley uses object-aware spatial controls [2], and Audible World Models anchors persistent sources in geometry [1]. Rotation equivariance and learned audio-visual evaluation also have direct precedents [6, 12]. Neither coordinate rotation nor the existence of a spatial generator or metric is a new contribution here.

At fixed sensor origins, two absolute orientations reduce to their relative rotation. Supplying two pose variables therefore does not itself introduce a new representation. The present contribution is a frozen diagnostic experiment: actual procedural waveforms, controlled relative-pose support, independent real source-position labels, two evaluation regimes and preserved negative controls. The question is whether these ingredients support a bounded synthetic-to-real component before testing a more ambitious data-allocation method.

The accompanying artifact provides code, frozen source identifiers, retrieval instructions, recorded predictions, small checkpoints and figure inputs at https://github.com/Saurav-K-Aryal/posesplit-pilot. Source recordings are retrieved separately; this manuscript contains numeric plots rather than photographs of participants.

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2 Methods

2.1 Real cohort and observation contract

STARSS23 supplies real spatial audio, synchronized panoramic video, human activity labels and source directions supported by optical tracking [5, 8]. We use its immutable version 1.1.0. Selection and source/code/protocol hashes are frozen before audio or video feature decoding and model training.

Eligibility uses only published metadata. A nonoverlapping half-second bin must contain exactly one annotated event in each of its five 10-Hz frames, with the same class and source throughout. Absolute elevation is at most 45 degrees and maximum pairwise directional motion is at most 15 degrees. A recording needs at least eight eligible bins and matching audio/video members. One annotated source does not exclude unannotated interference or establish audibility.

We choose the first eligible recording lexicographically in each of nine official development-training rooms, and two in each of seven official development-test rooms. Up to 12 eligible bins per recording are chosen by a fixed salted SHA-256 order, then stored chronologically. Every selected recording supplies 12 bins: 108 development windows and 168 locked windows. All headings and regimes derived from one window remain nested within its recording and room. Development data provide pipeline checks; no real audio or labels fit weights, select thresholds or tune features.

The target is source membership in a square 90-degree horizontal and vertical field of view. For unit camera-frame direction $d_c = (x, y, z)$,

$$V(d_c) = \mathbf{1}\{x > 0,\ |y| \le x + \varepsilon,\ |z| \le x + \varepsilon\}, \qquad \varepsilon = 10^{-9}. \tag{1}$$

This is geometric viewport membership, not unobstructed visibility, audibility or semantic object identity. The normalized mean of the five source-direction vectors supplies the grading target. Activity selection is externally labeled, so this is an oracle-activity spatial task rather than end-to-end event detection.

2.2 Coordinates, inputs and the two regimes

Audio uses 24-kHz, 16-bit, four-channel first-order Ambisonics (FOA), in ACN order W, Y, Z, X. Azimuth is positive toward the left, front is zero, and z points upward. With passive yaw rotation $R(-\psi)$, world direction d_w gives

$$d_m = R(-\psi_m)d_w, \qquad d_c = R(-(\psi_c - \psi_m))d_m = R(-\psi_c)d_w. \tag{2}$$

All training and evaluation use the same four relative headings: 0, 90, 180 and 270 degrees. No unseen-angle result is claimed.

In the *camera-only* regime, $\psi_m = 0$ and ψ_c takes those four values. The original decoded microphone PCM is identical across views. In the *virtual co-rig* regime, $\psi_m = \psi_c$; FOA is passively reoriented before feature extraction, and the relative camera/microphone yaw is zero. The panoramic camera crop at a given heading is shared by both regimes. This is an ideal reorientation of a recorded sound field, not physical microphone translation, a new acoustic room or a nonideal hardware-mount experiment.

A 512-sample Hann-window STFT with hop 256 retains 300–4,500 Hz. The intensity feature sums $\mathrm{Re}(W^*[X, Y, Z])$ over these time-frequency bins and normalizes its three-vector. A clipped coherence statistic divides its magnitude by the geometric mean of pressure and directional-channel energies. The six model inputs are normalized intensity (I_x, I_y, I_z), coherence, and $\cos\delta$, $\sin\delta$ for known relative yaw $\delta = \psi_c - \psi_m$. No source direction, event class, room identity, filename, target field-of-view label or image pixel enters the learned predictor.

Real video is used only for the media interface audit. Three frames from the first selected window of each recording are decoded into 320-by-320 perspective crops at the four headings. Pixel and PCM hashes preserve the relationship between regimes. This demonstrates construction of actual camera views; it does not validate a visual model or confirm an unoccluded source.

2.3 Synthetic waveforms and matched diagnostic arms

Seeds 101, 211, 307, 401 and 503 each produce 256 training and 64 validation waveforms. Thus the 1,600 total waveforms comprise 1,280 training and 320 validation examples. Each lasts half a second and is emitted as actual four-channel 16-bit WAV, then decoded before feature extraction.

A source combines band-limited colored noise, a tonal component and an amplitude envelope. A direct plane-wave FOA field is combined with six first-order rectangular-shoebox image-source reflections. Rooms vary from 4–8 m horizontally and 2.5–3.5 m vertically; source distance is .4–1.2 m, elevation −25 to 25 degrees, reflection coefficients .15–.65 and sensor noise 10–30 dB SNR. Propagation speed is 343 m/s. Channels are jointly scaled before PCM conversion. Source azimuths within one degree of a viewport boundary are excluded; each synthetic source belongs to exactly one of the four views. This deliberately limited model is not a validated full-wave simulator or semantically realistic household audio.

Every arm uses the same source-waveform inventory and 1,024 training rows per seed. The *pose-aware* arm pairs each waveform's features with all four known relative headings and their correct projection labels. The *canonical* arm repeats the same feature vector four times at relative yaw zero with its correct zero-yaw label. It has equal row budget but lacks independent relative-pose support. The *wrong-pose* arm has exactly the pose-aware arm's six-dimensional feature multiset, while a deterministic nonzero cyclic shift moves its four labels within each source block. This intentionally invalid-label arm is a diagnostic control, not a competitive augmentation baseline.

Each arm trains a six-input, 32-hidden-unit tanh MLP with a sigmoid output, giving 15 heads across five seeds. Initial weights match across arms within a seed. Full-batch binary cross-entropy uses Adam for 1,000 steps, learning rate .01, standard .9/.999 moments, numerical epsilon 10^{-8} and 10^{-4} L2 weight penalty. The last checkpoint is used, without early stopping or validation-driven retuning. A probability of at least .5 predicts membership. The pose-aware arm is uniform four-heading exposure; it is not the future targeted coverage sampler.

2.4 References, aggregation and independent replay

The analytic reference projects the measured intensity direction using known pose. A separate constant-front reference supplies a localization comparison. Both are evaluated against external source-position labels. The MLP predicts only binary membership and cannot be credited with improving continuous direction estimation.

The primary statistic pools confusion counts over locked camera-only cases within each seed, computes balanced accuracy, then averages the five seed scores. Each seed has 672 view cases, including 168 positives. A constant-class predictor has balanced accuracy .5, although always predicting negative would obtain .75 ordinary accuracy. We report the virtual co-rig regime alongside the primary regime.

A paired percentile bootstrap draws seven room clusters with replacement 2,000 times, using identical draws for all arms and seeds and preserving every repeated cluster's rows. It recomputes the same pooled-within-seed estimator. Undefined replicates would remain null without redraw; none occurs. Per-room and per-recording values are retained. These small-sample intervals are descriptive, conditional on the selected benchmark cohort, rather than guarantees about a device population.

Table 1. Locked geometric field-of-view balanced accuracy. Learned entries average five seed scores; the analytic entry has no training seed. Brackets are descriptive 95% room-cluster intervals over the same seven rooms.

Training/reference	Camera only	95% interval	Virtual co-rig	95% interval
Pose-aware	.829	[.735, .922]	.832	[.741, .924]
Canonical orientation	.480	[.464, .494]	.841	[.757, .929]
Wrong-pose labels	.408	[.374, .442]	.418	[.381, .452]
Analytic geometry	.782	[.678, .884]	.782	[.678, .884]

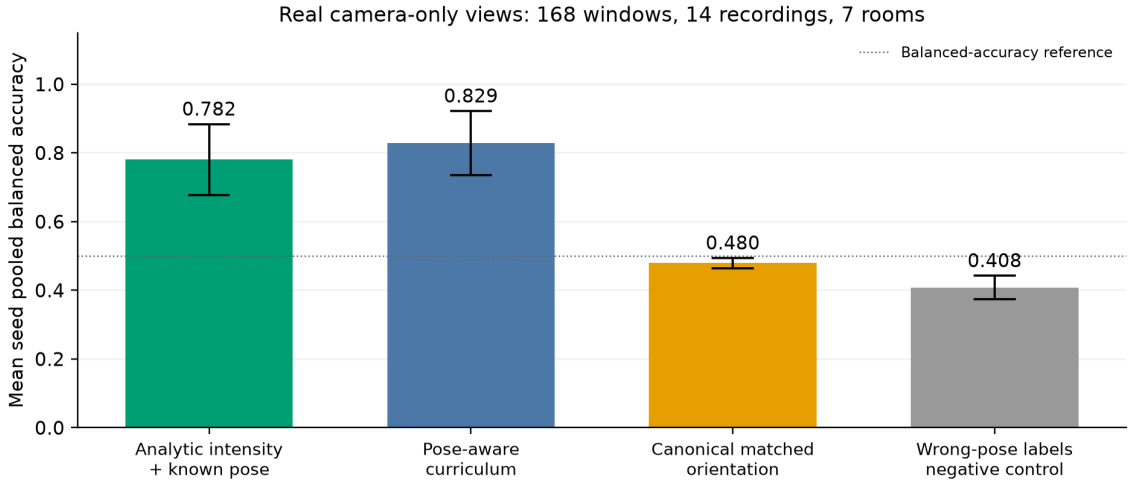


Fig. 1. Camera-only results. The four views reuse identical real microphone PCM. Canonical and incorrect-label arms are diagnostic controls; the co-rig counter-result is reported in Table 1.

Any source, feature or media decoding failure halts the study before fitting, preserving IDs and logs. All selected windows passed. An independent CPU readback verifies 46 source media members, 2,208 decoded real variants, 1,600 synthetic WAVs, 15 checkpoints, 35,328 recorded predictions across development and locked data, and all bootstrap replicates. The study uses no hosted model calls, replacement items or post-test tuning.

3 Results

3.1 Both mounting regimes matter

Table 1 gives both regimes. Camera-only pose-aware training reaches .829 mean balanced accuracy, compared with .480 canonical, .408 wrong-pose and .782 analytic. Figure 1 visualizes this primary condition. The pose-aware minus canonical difference is .349, with paired room-bootstrap interval [.252, .440]; the difference from deliberately wrong labels is .421 [.302, .543]. These large diagnostic contrasts show the consequence of missing pose support or invalid coupling. They do not establish superiority to uniform independent-pose augmentation.

The co-rig point estimates reverse the canonical comparison: .841 canonical versus .832 pose-aware. This regime has relative yaw zero, matching canonical training support. The result rules out an unqualified claim that the pose-aware curriculum dominates every mounting regime. Both regimes share the same recorded windows and are not independent real replications. Equal analytic scores reflect the coordinate transformation.

Table 2. Analytic direction estimates on 168 locked windows before repeated camera views. These are oracle-activity localization diagnostics, separate from learned binary membership.

Reference	Median error (deg)	Mean error (deg)	Within 20 deg
Intensity direction	23.48	30.16	.363
Constant front	50.14	58.10	.196

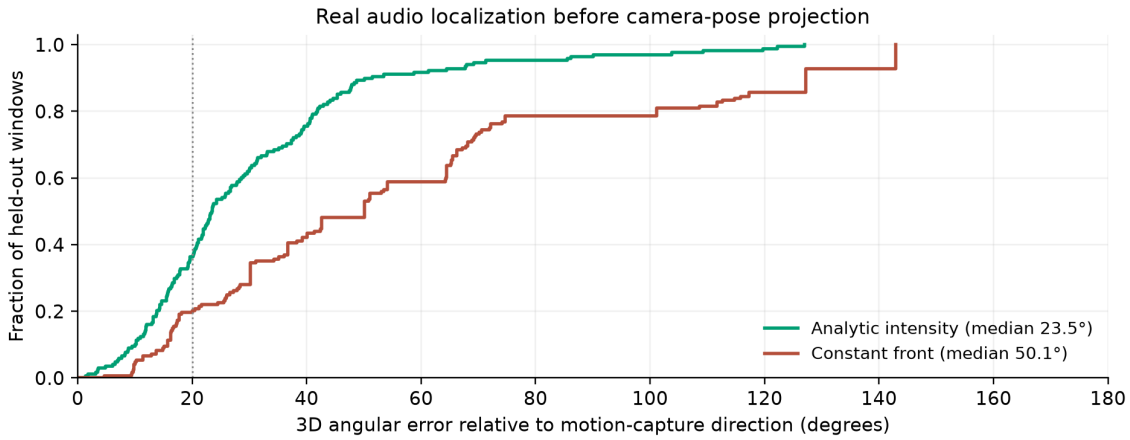


Fig. 2. Descriptive cumulative error distributions for the analytic and constant-front direction references. Repeated windows remain clustered within seven rooms.

The camera-only pose-aware minus analytic difference is .0476, with paired interval [.0067, .1038]. Six of seven rooms favor the learned binary head, but room 2 contributes about 59% of the average gain; one room slightly favors analytic geometry. The result is heterogeneous. Sensitivity rises from .631 analytic to .745 pose-aware, while specificity falls from .933 to .913. A threshold trade-off and this small cohort can explain an aggregate classification gain without implying a new localization method.

3.2 Measured spatial signal and resources

Table 2 and Figure 2 evaluate continuous direction from the analytic intensity feature. Median error is 23.48 degrees versus 50.14 for constant front, and the within-20-degree fraction is .363 versus .196. These measurements indicate useful spatial information in the real decoded audio. They are not outputs of the trained membership head.

The completed execution took 31.75 seconds after source retrieval and recorded approximately 349 MB peak Python RSS on an Apple M5 Max host with 128 GB RAM. Source retrieval and total-system or subprocess memory are not included in that RSS claim. This is a bounded CPU feasibility observation, not certification on an exact 48 GB laptop or a latency/energy benchmark for a deployed device.

3.3 What the pilot establishes

Actual procedural PCM training, externally labeled real recordings and grouped replay provide more than a synthetic self-check. They support an executable audio-plus-known-pose interface and show that training pose support matters in

the selected camera-only task. No real labels are used for fitting. The geometric measurements and media construction remain auditable even where classification fails.

The limits are substantial: one public dataset, seven locked rooms, short low-motion windows, a single annotated active source and four shared headings. Unknown extrinsics, continuous unseen yaw, translated microphones, nonideal arrays, physical occlusion and polyphonic source association remain untested. No image features enter the head. The plots therefore do not establish learned audio-visual fusion, source identity binding, human-perceived realism or a neural generator. Brier and log-loss diagnostics do not establish calibrated uncertainty, particularly when the analytic reference supplies hard zero/one outputs. Most importantly, a weak canonical arm and intentionally invalid labels cannot establish the proposed allocator's advantage over a strong independent-pose baseline.

4 Future Work

The next study should test a matched-budget coverage method, rather than treating two pose variables or coordinate rotation as novelty. A proposed one-year dataset targets 20,000 ten-second procedural clips across six to ten layouts, with at most 2,000 neural-refined clips. The scene specification should retain source identity, event timing, world trajectories, sensor transforms and acoustic assumptions. Physical translations require fresh propagation rendering; rotating an existing FOA field is insufficient.

The proposed allocator uses 48 strata: eight relative-yaw bins, three signed viewport-margin ranges, and dry versus reverberant regimes. Half of its allocation remains uniform over feasible strata. The remainder follows normalized development-only weights

$$w_b = e_b + \frac{1}{\sqrt{n_b + 1}}, \qquad e_b = \frac{k_b + 1}{m_b + 2}, \tag{3}$$

where n_b is training exposure and k_b errors occur among m_b development examples. Smoothed ordinary error is defined even when a stratum has only one target class. Source-event inventory, exposure and compute budgets must match uniform independent-pose augmentation; update checkpoints must freeze before locked evaluation. This allocator has not been implemented or tested in the present pilot.

A small frozen-backbone adapter can serve as a test vehicle. Comparisons should include available ViSAGe, PhysWave, StereoFoley and spatial-generation baselines while declaring missing implementations, format changes and extra supervision. Two fixed compact downstream backbones and three seeds should compare targeted versus uniform independent augmentation at an 8,000-training-clip primary operating point. The prospective useful-effect target is at least five balanced-accuracy percentage points, a paired room-bootstrap interval above zero, and no negative mean effect on either backbone. New development cluster variance should determine attainable precision; insufficient precision is inconclusive. A separate development gate of three points on each backbone should precede expansion of neural refinement. These thresholds are future design choices, not retrospective gates for this completed three-arm diagnostic.

The locked study should hold out mounts, rooms, source assets and trajectories, then examine genuinely unseen layouts and angles. Generated direction consistency, timing and perceptual quality need separate measurements, including a blinded human audit. Actual pixel features and source binding require their own comparisons before claiming visual complementarity. A small controlled capture study can test nonideal hardware when permissions and resources permit. A negative curriculum result should preserve the generator, matched learning curves and failure map rather than trigger expansion against the locked set.

5 Conclusion

A frozen pilot separates camera-only reframing from virtual co-rig reorientation using procedural spatial waveforms, known poses and independent real source-position labels. Pose-aware training reaches .829 balanced accuracy in the camera-only task, while canonical training is slightly higher in the co-rig condition. The analytic spatial baseline remains competitive. The evidence supports a bounded real-room evaluation component and identifies a clear next comparison: targeted coverage versus uniform independent-pose augmentation under matched budgets. It does not establish a new localizer, learned visual grounding or completed neural audio-visual generation.

Artifact preparation. Saurav K. Aryal directed the preliminary experiments with Codex assistance for implementation, execution, analysis and manuscript preparation. The public artifact retains frozen observations, source provenance and independent replay instructions.

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